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Loss Function & Backbone Ablation Study

Loss function comparison table:

Loss Function	mIoU	BG IoU	Intact IoU	Damaged IoU	Destroyed IoU
CE + Dice (baseline)	0.46				
Focal Tversky (0.3/0.7/2.0)	0.5181	0.947	0.537	0.330	0.257
Focal Tversky (0.25/0.75/1.33)	0.5203	0.945	0.529	0.333	0.2720
Phased (FT → +Lovász)	0.5233	0.950	0.537	0.332	0.2722

The phased compound loss strategy outperformed single-function approaches. Focal Tversky in early epochs (0–32) penalizes false negatives heavily ($\alpha=0.3$, $\beta=0.7$, $\gamma=2.0$), forcing the model to learn hard negatives and damaged/destroyed classes. Lovász loss injection at 80% training (epoch 32+) refines boundaries with differentiable IoU optimization, achieving the **highest mIoU and F1**. The improvement margin over C3 is modest (+0.003 mIoU), but F1 gain (+0.0021) and per-class stability suggest robustness.

Detailed metrics of the loss functions:

C2 Precision, Recall, F1[Loss Function- Focal Tversky (0.3/0.7/2.0)]:

Class	IoU	Precision	Recall	F1
Background	0.9479	0.9869	0.9600	0.9733
Intact	0.5370	0.6074	0.8225	0.6988
Damaged	0.3306	0.4442	0.5639	0.4970
Destroyed	0.2568	0.3685	0.4586	0.4086
Mean IoU: 0.5181				
Mean F1: 0.6444				

C3 Precision, Recall, F1[Loss Function- Focal Tversky (0.25/0.75/1.33)]:

Class	IoU	Precision	Recall	F1
Background	0.9460	0.9874	0.9576	0.9722
Intact	0.5292	0.5911	0.8347	0.6921
Damaged	0.3333	0.4704	0.5335	0.5000
Destroyed	0.2727	0.3723	0.5047	0.4285

Mean IoU: 0.5203

Mean F1: 0.6482

C4 Precision, Recall, F1[Loss Function- Phased (FT → +Lovász)]:

Class	IoU	Precision	Recall	F1
Background	0.9508	0.9839	0.9658	0.9748
Intact	0.5378	0.6280	0.7894	0.6995
Damaged	0.3326	0.4818	0.5178	0.4991
Destroyed	0.2722	0.4121	0.4451	0.4279

Mean IoU: 0.5233

Mean F1: 0.6503

****C2 Damaged (Recall=0.5639) and Destroyed (Recall=0.4586) classes show robust sensitivity—the model catches ~56% and ~45% of true positives respectively.

**** C4 has better F1 score (more balanced) and 0.5233 mIoU (+13.8% than baseline).

Backbone comparison table:

Backbone	mIoU	Damaged IoU	Destroyed IoU	Params	Train Time
ResNet-34 (baseline)	0.46			21.8M	
ResNet-50	0.5296	0.3372	0.2790	25.6M	2.9hrs
EfficientNet-B4	0.5257	0.3355	0.2725	19.3M	4.01hrs

ResNet-50 achieved the **highest mIoU of 0.5296 (+15.1% over baseline)**, with superior per-class performance on critical classes: **Damaged IoU = 0.3372, Destroyed IoU = 0.2790**. Extracted more discriminative features for rare classes without degrading speed (2.9 hrs).

Recommended Configuration

Encoder Backbone	ResNet-50 (25.6M params)
Loss Function	Phased Focal Tversky → Lovász $\alpha=0.3, \beta=0.7, \gamma=2.0$ (0–32 epochs) Switch to Lovász at 80% (epoch 32+) Weight: 0.6 Tversky, 0.4 Lovász
Expected mIoU	~0.53 (+15% over baseline)
Training Time	2.9+ hours (40 epochs, batch 16)
Critical Classes	Damaged IoU ≈ 0.33 Destroyed IoU ≈ 0.27

*****Note:** An Ablation Study with [Loss Function- Focal Tversky (0.3/0.7/2.0)] + ResNet-50 (25.6M params) can be done if Recall is the priority.